Mohamad Shafik bin Mohamad Adam

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SUMMARY

Software engineer building cloud services and semantic-metadata tooling with Python, TypeScript and AWS. Owns work end to end, from feasibility studies and architecture to APIs, testing and delivery. NUS Computer Science graduate.

TECHNICAL SKILLS

- Languages: Python, TypeScript, JavaScript, SQL; familiar with Java
- Frontend: Svelte, React, React Native, Streamlit, HTML/CSS
- Cloud and infrastructure: AWS (Lambda, AppSync, DynamoDB, Neptune, S3, CloudFront, CloudWatch), Docker, GitHub Actions
- Data and APIs: GraphQL, RDF, Brick Schema, SPARQL, graph databases, DynamoDB, Firebase
- Practices and tools: Git, Clean Architecture, pytest, MSW, Claude Code, GitHub Copilot

EXPERIENCE

Systems and Application Engineer, Software Development, Azbil ASPO, Singapore Jan 2026 to present

- Designing and building, as sole owner, a cloud metadata retrieval service on AWS Neptune and S3 whose APIs serve building metadata and point relationships to the building management platform.
- Ran a feasibility study on querying Brick Schema RDF data in AWS Neptune, documenting limits and trade-offs that guided architecture decisions, and evaluated PyOxigraph as an on-premises alternative.
- Designed and built an internal Streamlit app over 5 months that turns BMS point lists into Brick Schema metadata files through a guided workflow, completing the metadata pipeline that feeds the retrieval service.
- Delivered a cloud building management platform's event log feature end to end: a GraphQL API on AWS AppSync with cursor-based DynamoDB pagination (Clean Architecture) and a Svelte infinite-scroll table, tested with MSW mocks.
- Contributed to the platform MVP over 9 months in a 3-developer team (AWS Lambda, AppSync, DynamoDB, S3, CloudFront); leadership approved it for development toward commercial release.
- Present plans, progress and research findings to a partner R&D team monthly; proposed an LLM-based building health report MVP in the company's LLM study.

Software Engineer Intern, Azbil ASPO, Singapore Jun 2025 to Dec 2025

- Built a Python tool that parses Brick Schema TTL metadata and runs SPARQL queries to generate customer product configuration files, cutting preparation from about 3 days of manual engineering work to under 5 minutes.

Software Developer Intern, Ground Labs, Singapore May 2022 to Aug 2022

- Developed frontend features for a desktop data-discovery product, including a four-pane resizable window layout and match highlighting in search results.

PROJECTS

edge-energy-optimizer, building energy forecasting Oct 2026, in progress

- Built a day-ahead building load forecast with 34.5 kWh MAE, 24.5% better than a same-hour-last-week baseline.

EDUCATION

National University of Singapore, School of Computing graduated Jan 2026

- Bachelor of Computing (Computer Science), Honours (Merit)
- Relevant coursework: Software Engineering; Data Structures and Algorithms; Introduction to AI

Singapore Polytechnic Apr 2015 to May 2018

- Diploma in Electrical and Electronic Engineering (Computer and Power specialisation)

CERTIFICATIONS

- AWS Certified Cloud Practitioner (CLF-C02), in progress, expected November 2026